

WATCH TOGETHER

Real-Time Video Watching & Chat Application

Java • Spring Boot • MySQL • Hibernate/JPA • WebSocket • HTML/CSS/JavaScript

1. Project Overview

- Watch Together is a web application that allows users to watch videos together in a shared room.
- Users can register and log in to the application.
- Users can create private rooms or join existing rooms using a room code and password.
- Video actions such as Play, Pause and Seek are synchronized between users.
- Users can communicate through real-time chat inside the room.

2. Main Features

- User Registration and Login
- Create Room with room name and password
- Join, leave and rejoin rooms
- Real-time video synchronization using WebSocket/STOMP
- Real-time chat with sender and timestamp
- Chat messages are stored in MySQL
- User and room data managed using Hibernate/JPA and JDBC

3. Technology Stack & Architecture

- Frontend: HTML, CSS and JavaScript
- Backend: Java with Spring Boot
- Database: MySQL
- ORM: Hibernate / JPA
- Database Access: Spring Data JPA and JDBC
- Real-time communication: WebSocket with STOMP
- Flow: Frontend → REST API / WebSocket → Spring Boot → MySQL

4. Working Flow

1. User registers or logs in.
2. User reaches the dashboard.
3. User creates a room or joins using room code and password.
4. Users in the same room receive synchronized video actions.
5. Chat messages are sent through WebSocket and displayed instantly.
6. Chat history is loaded from MySQL when the room is opened.

5. Future Scope

- Support multiple video sources and a better video library.
- Add voice/video calling between room members.
- Add user profiles, friends and invitations.
- Use BCrypt password hashing and secure authentication.
- Deploy the application online.
- Improve notifications, responsive UI and room controls.